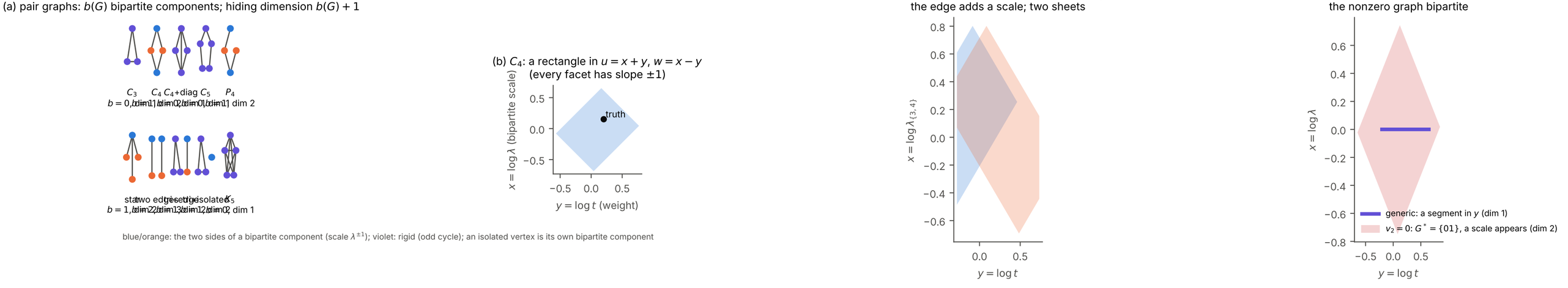
(a) pair graphs: $b(G)$ bipartite components; hiding dimension $b(G)+1$
C_3
$b=0$, dim 1
C_4
$b=1$, dim 2
C_4+diag
$b=0$, dim 1
C_5
$b=1$, dim 2
P_4
$b=1$, dim 2
star
$b=1$, dim 2
two edges
$b=1$, dim 2
tree
$b=1$, dim 2
three edges
$b=1$, dim 2
isolated K_5
$b=2$, dim 1
(b) C_4: a rectangle in $u=x+y$, $w=x-y$
(every facet has slope ±1)
truth
$x=\log\lambda$ (bipartite scale)
$y=\log t$ (weight)
blue/orange: the two sides of a bipartite component (scale $\lambda^{\pm1}$); violet: rigid (odd cycle); an isolated vertex is its own bipartite component
(c) triangle + edge: the triangle bounds t,
the edge adds a scale; two sheets
$x=\log\lambda_{\{3,4\}}$
$y=\log t$
sheet $(+,+)$
sheet $(+,-)$ (labels swapped on $\{3,4\}$)
(d) the zero stratum: a coincidence makes
the nonzero graph bipartite
$x=\log\lambda$
$y=\log t$
generic: a segment in y (dim 1)
$v_2=0$: $G^*=\{01\}$, a scale appears (dim 2)